Ellie Paek

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Education

Emory University Aug 2021 – May 2024
Bachelor of Science in Mathematics and Computer Science with Highest Honors (GPA: 3.95) *Atlanta, Georgia*
- **Relevant Coursework:** Data Structures & Algorithms, Systems Programming, Machine Learning
- **Thesis:** Emora Assistant Bot: Revolutionizing Task Automation and Chatbot Evaluation

Emory University Aug 2021 – May 2024
Bachelor of Arts in Music, Vocal Performance *Atlanta, Georgia*

Experience

Emory University School of Nursing Feb 2026 – Jul 2026
Software Engineer (Part-Time) *Remote*
- Built a speech-enabled AI roleplay agent using **Python, LangChain, and OpenAI APIs**, integrating **STT/TTS** for **100+** users.
- Optimized real-time dialogue flows, improving turn-taking and response latency.
- Developed a **Django** platform to deploy and manage concurrent sessions.

Emory NLP Research Lab Jul 2024 – Sep 2025
AI Research Scientist *Atlanta, GA*
- Fine-tuned multilingual encoders for RAG systems, improving cross-lingual retrieval accuracy to **70%**.
- Built and evaluated dialect-aware chatbots, achieving **91%** feature accuracy.
- Designed evaluation workflows for multilingual model outputs.

Emory NLP Research Lab May 2022 – May 2024
Undergraduate Researcher *Atlanta, GA*
- Developed a multi-party dialogue system with LLM-based evaluation, achieving **88%** task completion and reducing manual evaluation by **90%**.

Emory University School of Medicine Oct 2022 – May 2024
XR Developer (Intern) *Atlanta, GA*
- Developed VR-based cognitive assessment simulations for epileptic patients using **Unity and C#**, designing immersive environments for neurological testing.

Projects

Musaic | *Python, React.js, OpenAI* Jan 2023 – May 2023
- Built a web app converting natural language prompts into Spotify playlists using **OpenAI API**.

POVTales | *Python, Streamlit, OpenAI API, RAG*
- Built a **retrieval-augmented** chatbot that enables conversations with literary characters while grounding responses in story canon and character perspective.
- Implemented **timeline-aware retrieval**, **character knowledge modeling**, and **automated evaluation pipelines** to improve response consistency and quality.

Selected Publications (6 total)

- "Finding A Voice: Evaluating African American Dialect Generation for Chatbot Technology," ACL 2025.
- "Reference-Aligned Retrieval-Augmented Question Answering over Heterogeneous Proprietary Documents," CIKM 2025 (Applied Research Track).
- "Enhancing Task-Oriented Dialogue Systems through Synchronous Multi-Party Interaction," Information, 2024.

Technical Skills

Languages: Python, TypeScript, JavaScript, C++, Java, C#, SQL
Frameworks: Django, Flask, Next.js, React, Tailwind CSS, Streamlit, Unity
AI/ML: PyTorch, HuggingFace, Scikit-learn, LangChain, FAISS, RAG pipelines, NLP, LLM evaluation, conversational AI systems
Cloud & Tools: AWS (S3, Lex, Bedrock), Docker, Git, Linux